

Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

The below Terraform file creates Amazon Linux EC2 instances in Mumbai (ap-south-1) and Singapore (ap-southeast-1) and outputs both public IPs.

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  alias  = "mumbai"
  region = "ap-south-1"
}

provider "aws" {
  alias  = "singapore"
  region = "ap-southeast-1"
}

data "aws_ami" "mumbai_server" {
  provider      = aws.mumbai
  most_recent   = true
  owners        = ["amazon"]

  filter {
    name     = "name"
    values   = ["al2023-ami-*-x86_64"]
  }
}

data "aws_ami" "singapore_server" {
  provider      = aws.singapore
  most_recent   = true
  owners        = ["amazon"]

  filter {
```

```

    name      = "name"
    values    = ["a12023-ami-*-x86_64"]
  }
}

resource "aws_instance" "mumbai" {
  provider      = aws.mumbai
  ami           = data.aws_ami.mumbai_server.id
  instance_type = "t3.micro"

  tags = {
    Name = "Mumbai-server"
  }
}

resource "aws_instance" "singapore" {
  provider      = aws.singapore
  ami           = data.aws_ami.singapore_server.id
  instance_type = "t3.micro"

  tags = {
    Name = "Singapore-server"
  }
}

output "public_ips" {
  value = {
    mumbai    = aws_instance.mumbai.public_ip
    singapore = aws_instance.singapore.public_ip
  }
}

```

Steps to get the output:

1. Terraform init

```

● sathish@Jeeva-M1 single-file-ec2 % terraform init
  Initializing provider plugins found in the configuration...
  - Finding hashicorp/aws versions matching "~> 5.0"...
  - Installing hashicorp/aws v5.100.0...
  - Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

  Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

```

2. Terraform validate

```

● sathish@Jeeva-M1 single-file-ec2 % terraform validate
Success! The configuration is valid.

```

3. Terraform plan

```

● sathish@Jeeva-M1 single-file-ec2 % terraform plan
data.aws_ami.singapore_server: Reading...
data.aws_ami.mumbai_server: Reading...
data.aws_ami.mumbai_server: Read complete after 1s [id=ami-0eef9d717347382c1]
data.aws_ami.singapore_server: Read complete after 1s [id=ami-010ead48412d6f271]

```

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ public_ips = {
  + mumbai      = (known after apply)
  + singapore   = (known after apply)
}

```

4. Terraform apply

```

sathish@Jeeva-M1 single-file-ec2 % terraform apply --auto-approve
data.aws_ami.singapore_server: Reading...
data.aws_ami.mumbai_server: Reading...
data.aws_ami.mumbai_server: Read complete after 1s [id=ami-0eef9d717347382c1]
data.aws_ami.singapore_server: Read complete after 1s [id=ami-010ead48412d6f271]

aws_instance.mumbai: Creating...
aws_instance.singapore: Creating...
aws_instance.mumbai: Still creating... [00m10s elapsed]
aws_instance.singapore: Still creating... [00m10s elapsed]
aws_instance.mumbai: Creation complete after 12s [id=i-05a92d3a65c3fc4a2]
aws_instance.singapore: Creation complete after 13s [id=i-05d8cc454ff46da69]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

public_ips = {
  "mumbai" = "35.154.98.29"
  "singapore" = "13.229.149.133"
}
sathish@Jeeva-M1 single-file-ec2 %

```

Screenshot of Mumbai instance in AWS console:

Instance summary for i-05a92d3a65c3fc4a2 (Mumbai-server) [Info](#)

Updated less than a minute ago

Instance ID i-05a92d3a65c3fc4a2	Public IPv4 address 35.154.98.29 open address	Private IPv4 addresses 172.31.41.49
IPv6 address -	Instance state Running	Public DNS -

Screenshot of the Singapore instance in the AWS console:

Instance summary for i-05d8cc454ff46da69 (Singapore-server) [Info](#)

Updated less than a minute ago

Instance ID i-05d8cc454ff46da69	Public IPv4 address 13.229.149.133 open address	Private IPv4 addresses 172.31.10.61
IPv6 address -	Instance state Running	Public DNS -